

● RAG Learning

1.3 Measures, Accuracy & Bounds

Pearson Edexcel GCSE Mathematics 1MA1 — Number — spec refs N13–N16 — Papers 1 (non-calculator), 2 and 3 (calculator)

1.3.1 Standard Units and Compound Measures (N13)

Edexcel N13 asks you to use standard units of **mass, length, time, money and other measures** — including standard compound measures such as speed, density and pressure — using decimal quantities where appropriate. Metric conversion factors are not given on the paper, so you must **know** them. Conversions between metric and imperial units (for example 5 miles $\approx$ 8 km) are always **given in the question**.

Measure	Conversions you must know
Length	10 mm = 1 cm 100 cm = 1 m 1000 m = 1 km
Mass	1000 mg = 1 g 1000 g = 1 kg 1000 kg = 1 tonne
Capacity	1000 ml = 1 litre 100 cl = 1 litre 1 ml = 1 cm ³ 1000 cm ³ = 1 litre
Time	60 s = 1 min 60 min = 1 hour 24 h = 1 day 7 days = 1 week 365 days = 1 year (366 in a leap year)
Money	100p = £1

Which way? Changing to a *smaller* unit gives a *bigger* number, so **multiply**. Changing to a *larger* unit gives a smaller number, so **divide**. A quick sense check: there are more centimetres than metres in any length.

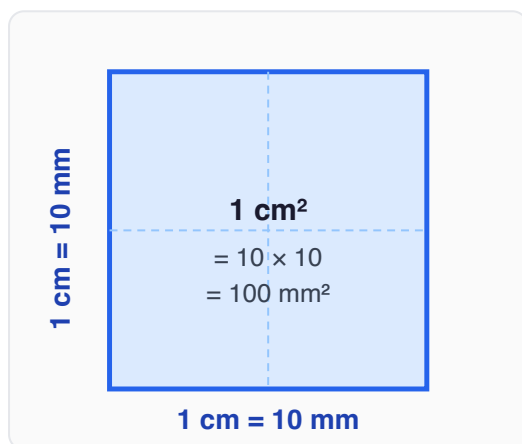
Worked Example: Change 0.86 km into centimetres.

$$0.86 \text{ km} = 0.86 \times 1000 = 860 \text{ m}$$

$$860 \text{ m} = 860 \times 100 = \mathbf{86\,000 \text{ cm}}$$

Area and Volume Units

An area unit is a square, so the length factor is applied **twice** (squared). A volume unit is a cube, so the length factor is applied **three times** (cubed).



Area

$$1 \text{ cm}^2 = 10^2 = 100 \text{ mm}^2$$

$$1 \text{ m}^2 = 100^2 = 10\,000 \text{ cm}^2$$

Volume and capacity

$$1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$$

$$1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$$

$$1 \text{ m}^3 = 1000 \text{ litres}, \quad 1 \text{ litre} = 1000 \text{ cm}^3$$

Worked Example: Change 3.7 m² into cm².

$$3.7 \times 100 \times 100 = 3.7 \times 10\,000 = \mathbf{37\,000 \text{ cm}^2}$$

Worked Example: A water butt holds 2.5 m³. How many litres is this? How many 750 cm³ jugs could be filled from 3 litres?

$$2.5 \times 1000 = \mathbf{2500 \text{ litres}}$$

$$3 \text{ litres} = 3000 \text{ cm}^3, \text{ and } 3000 \div 750 = \mathbf{4 \text{ jugs}}$$

Exam Tip: Do not convert m² to cm² by multiplying by 100. Picture the 1 m by 1 m square: it is 100 cm by 100 cm, so it contains 10 000 cm². The same trap appears with m³ (factor 1 000)

000, not 100).

Time and Compound Measures

Time is **base 60**, not base 10. 2.15 hours is *not* 2 hours 15 minutes: $0.15 \times 60 = 9$, so it is 2 hours 9 minutes. To turn minutes into a decimal of an hour, divide by 60.

$$\text{speed} = \frac{\text{distance}}{\text{time}} \quad \text{density} = \frac{\text{mass}}{\text{volume}} \quad \text{pressure} = \frac{\text{force}}{\text{area}}$$

Units follow the formula: $\text{km} \div \text{hours}$ gives km/h ; $\text{g} \div \text{cm}^3$ gives g/cm^3 ; $\text{N} \div \text{m}^2$ gives N/m^2 .

Worked Example: A lorry travels 162 km in 2 hours 15 minutes. Work out its average speed.

15 min = $15 \div 60 = 0.25$ h, so the time is 2.25 h.

speed = $162 \div 2.25 = \mathbf{72}$ km/h

Worked Example: A metal block has mass 540 g and volume 200 cm^3 . Work out its density.

density = $540 \div 200 = \mathbf{2.7}$ g/cm³

1.3.2 Rounding and Truncating (N15)

You must round numbers and measures to an **appropriate degree of accuracy** — to a given number of decimal places (d.p.) or significant figures (s.f.) — and decide on sensible accuracy in a context.

How to round: locate the last digit you will keep and look at the digit straight after it (the "decider").

- Decider 5, 6, 7, 8 or 9 → the kept digit goes **up** by one

- Decider 0, 1, 2, 3 or 4 → the kept digit **stays** the same

Decimal Places

Count the digits **after the decimal point**. Keep trailing zeros that are part of the required accuracy (4.9973 to 2 d.p. is 5.00, not 5).

Number	1 d.p.	2 d.p.	3 d.p.
6.4718	6.5	6.47	6.472
0.0839	0.1	0.08	0.084
4.9973	5.0	5.00	4.997

Significant Figures

Start counting at the **first non-zero digit**. Zeros in front of it are not significant; zeros between or after non-zero digits (in a decimal) are. When rounding a large number, use **zeros as place holders** so the number keeps its size.

0.04070 has 4 significant figures

0 . 0 4 0 7 0

leading zeros: not significant 4, 0, 7, 0 are significant

Number	1 s.f.	2 s.f.	3 s.f.
61 482	60 000	61 000	61 500
0.007351	0.007	0.0074	0.00735
2.0048	2	2.0	2.00

Worked Example: Round 0.0040519 to 3 significant figures.

The significant figures are 4, 0, 5, then the decider is 1. That is less than 5, so the 5 stays.

Answer: 0.00405

Truncation

To **truncate** a number, chop it off at the required place and ignore everything after — no rounding up ever happens. 5.678 truncated to 1 d.p. is 5.6 (rounded to 1 d.p. it would be 5.7).

Sensible Accuracy in Context

- Money is written to 2 d.p.: £8.5 should be written £8.50.
- Let the context decide the direction: 70 people travelling in 16-seat minibuses need $70 \div 16 = 4.375 \Rightarrow$ **5 minibuses**, but £70 buys only **4 tickets** at £16 each.
- Do not give an answer more accurately than the measurements it came from; matching the least accurate value used is a good rule.

Exam Tip: On the calculator papers, keep the full value on your calculator throughout a multi-step question and round only at the very end. Rounding part-way can push the final answer outside the accepted range.

1.3.3 Estimating and Checking (N14)

N14 asks you to **estimate answers** and to **check calculations** using approximation and estimation, including answers obtained using technology. These questions usually appear on Paper 1 (non-calculator).

Method: round each number to **1 significant figure** unless the question says otherwise, write the rounded values down, then work out the simpler calculation. The rounded values earn the method mark.

Worked Example: Work out an estimate for $\frac{7.83 \times 41.2}{0.19}$.

$$7.83 \approx 8, \quad 41.2 \approx 40, \quad 0.19 \approx 0.2$$

$$\frac{8 \times 40}{0.2} = \frac{320}{0.2} = \mathbf{1600} \quad (\text{the exact value is about 1698})$$

Worked Example: Zara uses her calculator to work out 52.3×0.187 and gets 97.801. Use estimation to check her answer.

Estimate: $50 \times 0.2 = 10$. Her answer is about 10 times too big, so it must be wrong (the correct value is 9.7801).

Is the Estimate Too Big or Too Small?

- Rounding a number you **add or multiply up** makes the estimate an overestimate.
- Rounding a number you **divide by up** makes the estimate an underestimate (you divide by more).
- If the roundings pull in different directions you cannot tell straight away — you need to compare with the exact answer.

Exam Tip: Dividing by 0.2 is the same as multiplying by 5: $320 \div 0.2 = 1600$. If you get a smaller answer when dividing by a number less than 1, something has gone wrong.

Estimating with Powers and Roots

Worked Example: Work out an estimate for $\frac{\sqrt{80.6} \times 2.1^3}{0.52}$.

$$\sqrt{80.6} \approx \sqrt{81} = 9, \quad 2.1^3 \approx 2^3 = 8, \quad 0.52 \approx 0.5$$

$$\frac{9 \times 8}{0.5} = \frac{72}{0.5} = \mathbf{144}$$

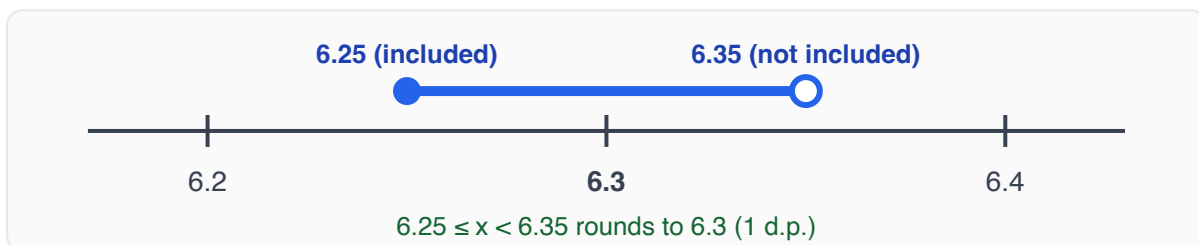
1.3.4 Error Intervals (N15)

A rounded or truncated value hides the true value somewhere in a range. Edexcel asks you to write this range, the **error interval**, using inequality notation.

If a value has been **rounded** to a certain unit, the true value is within **half a unit** of it:

$$\text{value} - \frac{1}{2}\text{unit} \leq x < \text{value} + \frac{1}{2}\text{unit}$$

Use $\leq$ at the bottom (that value rounds up to the given value) and $<$ at the top (that value would round up to the next one).



Worked Example: The mass w of a bag of sand is 23 kg, correct to the nearest kilogram. Write down the error interval for w .

Half of 1 kg is 0.5 kg: **$22.5 \leq w < 23.5$**

Worked Example: $n = 5800$, correct to 2 significant figures. Write down the error interval for n .

The second significant figure is the hundreds digit, so this is "to the nearest 100". Half of 100 is 50: **$5750 \leq n < 5850$**

Error Intervals after Truncation

A **truncated** value can only have lost digits, never gained them, so the true value runs from the stated value up to (but not including) the next value at that accuracy:

$$x = 9.2 \text{ truncated to 1 d.p.} \Rightarrow 9.2 \leq x < 9.3$$

Worked Example: A number n is 3.4 when rounded to 1 d.p., but 3.3 when truncated to 1 d.p. Find the error interval for n .

Rounding gives $3.35 \leq n < 3.45$. Truncation gives $3.3 \leq n < 3.4$.

Both must hold, so $3.35 \leq n < 3.4$

Exam Tip: Write the upper limit as 23.5, not 23.49 or 23.4999... The strict "<" sign is what shows the limit itself is excluded.

1.3.5 Limits of Accuracy and Bounds (N16)

The **lower bound** (LB) and **upper bound** (UB) of a measurement are the ends of its error interval — the least and greatest values it could have had before rounding. Interpreting limits of accuracy for a single value is on both tiers.

Measurement	Accuracy	Lower bound	Upper bound
40 kg	nearest 10 kg	35 kg	45 kg
8.6 s	nearest 0.1 s	8.55 s	8.65 s
150 m	nearest 5 m	147.5 m	152.5 m
3.72 cm	2 d.p.	3.715 cm	3.725 cm

Calculating with Upper and Lower Bounds Higher

The 1MA1 Higher statement of N16 adds "including upper and lower bounds": you combine bounds in calculations. Choose, for each value, the bound that pushes the answer as **high** as possible (for the UB) or as **low** as possible (for the LB).

Calculation	Upper bound	Lower bound
$a + b$	$UB_a + UB_b$	$LB_a + LB_b$
$a - b$	$UB_a - LB_b$	$LB_a - UB_b$
$a \times b$	$UB_a \times UB_b$	$LB_a \times LB_b$
$a \div b$	$UB_a \div LB_b$	$LB_a \div UB_b$

Subtracting or dividing by a *smaller* number gives a *larger* answer, so the second value takes the **opposite** bound.

Worked Example: A rectangle measures 7.3 cm by 2.6 cm, both correct to 1 d.p. Work out the lower bound of its area.

$$LB = 7.25 \times 2.55 = \mathbf{18.4875 \text{ cm}^2}$$

Worked Example: $a = 15.4$ and $b = 6.8$, both correct to 1 d.p. Work out the upper bound of $a - b$.

$$UB_a - LB_b = 15.45 - 6.75 = \mathbf{8.7}$$

Worked Example: A cyclist rides 400 m, correct to the nearest 10 m, in 61.2 s, correct to 1 d.p. Work out the upper bound of the average speed.

$$UB \text{ of speed} = \frac{UB \text{ distance}}{LB \text{ time}} = \frac{405}{61.15} = 6.623 \dots \approx \mathbf{6.62 \text{ m/s (3 s.f.)}}$$

Choosing a Suitable Degree of Accuracy Higher

Find *both* bounds and round them. The answer is given to the greatest accuracy at which the two bounds **round to the same value**; state this as your reason.

Worked Example: $x = 8.7$ and $y = 3.1$, both correct to 1 d.p. $z = x \div y$. Work out z to a suitable degree of accuracy.

$$\text{UB} = 8.75 \div 3.05 = 2.868 \dots \quad \text{LB} = 8.65 \div 3.15 = 2.746 \dots$$

To 2 s.f.: 2.9 and 2.7 — they differ. To 1 s.f.: 3 and 3 — they agree.

So $z = \mathbf{3}$ (1 s.f.), because both bounds round to 3 to 1 s.f.

Worked Example: A square has area 30 cm^2 , correct to the nearest cm^2 . Work out the lower bound of its side length.

$$\text{LB of area} = 29.5, \text{ so LB of side} = \sqrt{29.5} = 5.431 \dots \approx \mathbf{5.43 \text{ cm}}$$

Exam Tip: Write the bounds you are using before you calculate (e.g. "UB distance 405, LB time 61.15"). Each correct bound usually scores a B1 mark even if the final calculation goes wrong. Never round the bounds themselves.

Summary

Skill	Key idea
Units and compound measures (N13)	To a smaller unit: multiply. Square the factor for area, cube it for volume. Time is base 60.
Rounding (N15)	Look at the decider digit; 5 or more rounds up. Keep zeros as place holders.
Estimation (N14)	Round each number to 1 s.f., write the values down, then calculate.
Error interval — rounding (N15)	$\text{value} - \text{half unit} \leq x < \text{value} + \text{half unit}$

Error interval — truncation (N15)	value $\leq x <$ value + one unit
Bounds in calculations (N16, Higher)	Pick the bounds that make the result largest/smallest; use the opposite bound for $-$ and $\div$.